



Telecom Customer Churn Analysis

Project Documentation

Track: Data Analysis Specialist

Team Members	Instructor
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Abstract

This project presents an end-to-end data analysis of customer churn behavior in the telecommunications industry, conducted by a team of five data analysts as part of the Digital Egypt Pioneers Initiative (DEPI) — Data Analysis Specialist Track. Using a real-world dataset of 7,043 customer records originally published by IBM, the project explores the key drivers behind customer churn across demographic, contractual, and service-related dimensions. The analysis was performed across five industry-standard platforms — Microsoft Excel, SQL, Python, Power BI, and Tableau — each contributing a distinct analytical perspective to the same underlying data.

The project encompasses the full data analysis pipeline: data collection and merging, preprocessing and cleaning, star schema data modeling, exploratory and statistical analysis, machine learning prediction using a Random Forest classifier, and interactive dashboard development. Key findings reveal a 26.5% overall churn rate, with Month-to-Month contract customers churning at 45.8%, first-year customers at 47.4%, and senior citizens at 41.7%. The project concludes with actionable business recommendations targeting retention strategy, pricing competitiveness, and customer onboarding — grounded in both statistical evidence and machine learning insights.

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Chapter 1

Introduction

1.1 Project Overview

This project presents a comprehensive multi-tool analysis of customer churn behavior within the telecommunications industry. Using a real-world dataset originally published by IBM, the team performed end-to-end data analysis across five platforms: Microsoft Excel, SQL, Python, Power BI, and Tableau — each offering a unique analytical lens on the same underlying data.

The dataset encompasses 7,043 customer records with 38 attributes covering demographics, subscribed services, contract details, billing information, and churn outcomes. A data preprocessing phase involved merging two complementary datasets, cleaning inconsistencies, and structuring the data into a Star Schema data model consisting of one Fact table and four Dimension tables.

1.2 Motivation

Customer churn — the phenomenon where subscribers discontinue their service — is one of the most critical challenges facing the telecommunications industry. Acquiring a new customer costs five to ten times more than retaining an existing one, making churn prediction and prevention a high-priority strategic concern.

26.5% Overall Churn Rate	\$3.68M Revenue Lost to Churn	45.8% Month-to-Month Churn
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These figures underscore the business urgency of this project. By applying descriptive, diagnostic, and inferential analytics, the team aims to identify the root causes of churn, quantify the financial impact, and deliver actionable recommendations that can directly reduce churn rates.

1.3 Objectives

This project is designed to achieve the following objectives:

Build a robust data model	Design and implement a Star Schema (Fact + 4 Dimension tables) to enable efficient multi-dimensional analysis.
Perform descriptive analysis	Summarize key metrics including churn rate, average monthly charges, tenure distribution, and revenue breakdown across customer segments.
Identify churn drivers	Determine which contract types, service combinations, demographic groups, and payment methods are most strongly associated with churn.
Apply inferential statistics	Validate findings using T-Tests, Chi-Square tests, ANOVA, and regression analysis to confirm that observed patterns are statistically significant.
Deliver visual insights	Create interactive dashboards in Excel, Power BI, and Tableau that communicate findings clearly to both technical and non-technical stakeholders.
Provide actionable recommendations	Translate analytical findings into concrete business strategies targeting retention, pricing, and customer engagement.

1.4 Challenges

The team encountered and resolved several challenges throughout the project lifecycle:

- **Data Integration:** Merging two datasets from different sources required careful alignment of column structures, handling of duplicate records, and resolution of naming inconsistencies across 38 attributes.
- **Data Model Design:** Determining the optimal Star Schema structure involved critical decisions around which attributes belong in the Fact table versus Dimension tables — particularly for customer attributes and churn reasons, which required understanding Kimball methodology trade-offs.
- **Relationship Cardinality:** Excel Power Pivot enforced 1-to-many constraints that required rethinking the customer dimension approach, leading to a deliberate architectural decision to retain customer attributes in the Fact table.
- **Statistical Interpretation:** Translating p-values and test statistics from T-Tests, Chi-Square, and ANOVA into plain business language required bridging the gap between statistical rigor and managerial relevance.
- **Cross-Tool Consistency:** Ensuring that insights derived from Excel, SQL, Python, Power BI, and Tableau remained consistent and complementary across all five platforms presented an ongoing coordination challenge.

Chapter 2

Methodology

2.1 Dataset Description

The analysis is based on two complementary datasets sourced from Kaggle, both originating from IBM's publicly available telecom customer data. The datasets were merged to form a single unified dataset of 7,043 unique customer records spanning 38 attributes. These attributes cover four main categories:

Demographics: Age, gender, marital status, number of dependents, and senior citizen status.

Services: Phone service, internet service type, online security, online backup, device protection, premium tech support, streaming TV, streaming movies, streaming music, and unlimited data.

Contract & Billing: Contract type, payment method, paperless billing, monthly charges, total charges, total revenue, refunds, and extra data charges.

Churn Information: Churn status (Yes/No), churn category, and churn reason.

2.2 Data Sources

Dataset 1 — Telco Customer Churn (IBM/Kaggle)

Source: <https://www.kaggle.com/datasets/blatchar/telco-customer-churn>

Description: The original IBM telecom dataset containing 7,043 records and core customer attributes including demographics, subscribed services, contract information, and churn status.

Dataset 2 — Telecom Customer Churn by Maven Analytics

Source: <https://www.kaggle.com/datasets/shilongzhuang/telecom-customer-churn-by-maven-analytics>

Description: An enriched version of the same IBM dataset with additional columns including churn category, churn reason, total revenue, refunds, and extra data charges. Used to extend the analytical depth of the original dataset.

The two datasets were joined on the shared key `customer_id` using a `LEFT JOIN` to produce the final merged dataset: `telecom_merged`.

2.3 Data Preprocessing

Preprocessing was applied to both datasets individually before merging, and then to the merged dataset as a whole. The following steps were performed:

Step 1 — Data Merging

The two datasets were merged on `customer_id` using a `LEFT JOIN` to ensure all records from Dataset 1 were retained, with enriched columns from Dataset 2 appended where available.

Step 2 — Duplicate Removal

All duplicate rows were identified and removed from both datasets prior to merging. The final merged dataset contains 7,043 unique customer records with no duplicate customer IDs.

Step 3 — Null Value Handling

Null values were treated using context-aware logic rather than simple deletion or mean imputation:

Columns related to internet services (e.g., Online Security, Online Backup, Internet Type) were filled with "No internet service" for customers where Internet Service = No.

Columns related to phone services (e.g., Multiple Lines, Average Monthly Long Distance Charges) were filled with "No phone service" or 0 where Phone Service = No.

Churn Category and Churn Reason were filled with "No churn" for retained customers (Churn = No).

Remaining nulls in numeric columns (e.g., Avg Monthly GB Download) were filled with 0.

Step 4 — Data Type Standardization

`monthly_charges`, `total_charges`, and `Total_Revenue` were cast to `DECIMAL(10,2)` in SQL and `float64` in Python to ensure numeric precision.

`SeniorCitizen` was retained as integer (0/1) for ML modeling and mapped to "Senior" / "Non-Senior" labels for visualization.

`Zip_Code` was treated as TEXT to preserve leading zeros.

Step 5 — Value Standardization

The Married column values were standardized from "Yes" / "No" to "Married" / "Not Married" for clarity in visualizations.

Whitespace and inconsistent casing in categorical columns (e.g., Contract, Payment Method, Internet Type) were trimmed and normalized.

Step 6 — Feature Engineering

Three derived columns were created to support segmentation analysis:

Tenure_Group: Customers grouped into tenure buckets — 0–12 months, 13–24 months, 25–48 months, and 49–72 months — using CASE WHEN logic in SQL and pd.cut() in Python.

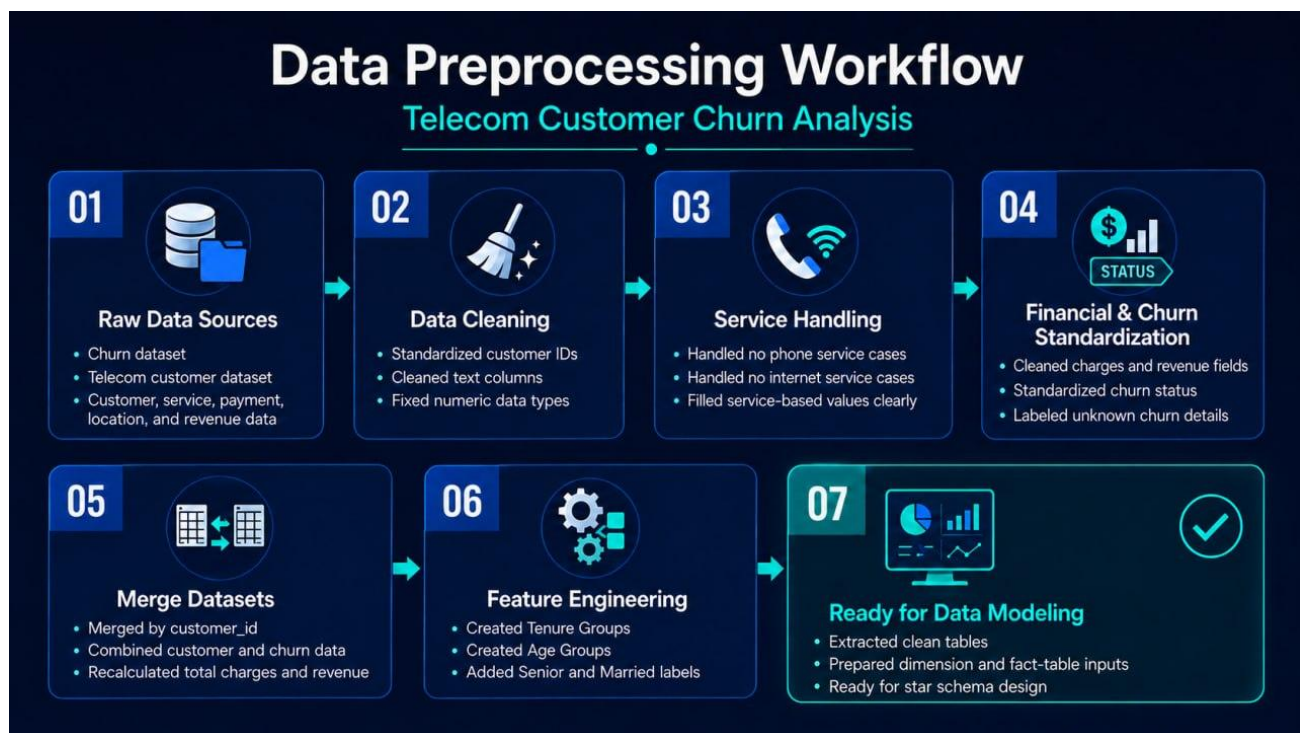
Age_Group: Customers grouped into age brackets — 18–30, 31–45, 46–60, and 60+ — using the same approach.

Senior_Label: A human-readable label derived from the binary SeniorCitizen column (0 → "Non-Senior", 1 → "Senior").

Step 7 — Total Revenue Calculation

In cases where Total Revenue was not directly available, it was calculated as:

$$\text{Total Revenue} = (\text{Monthly Charges} \times \text{Tenure}) + \text{Total Extra Data Charges} + \text{Total Long Distance Charges} - \text{Total Refunds}$$



Step 8 — Data Modeling — Star Schema

Following preprocessing, the data was restructured into a Star Schema consisting of one Fact table and five Dimension tables:

Dim_City: City, Zip Code, Latitude, Longitude

Dim_Contract: Contract type, Payment Method, Paperless Billing, Offer

Dim_Services: All 12 service-related attributes

Dim_Churn: Churn Category and Churn Reason

Dim_Customer: Customer demographics (gender, age, marital status, dependents, senior citizen status)

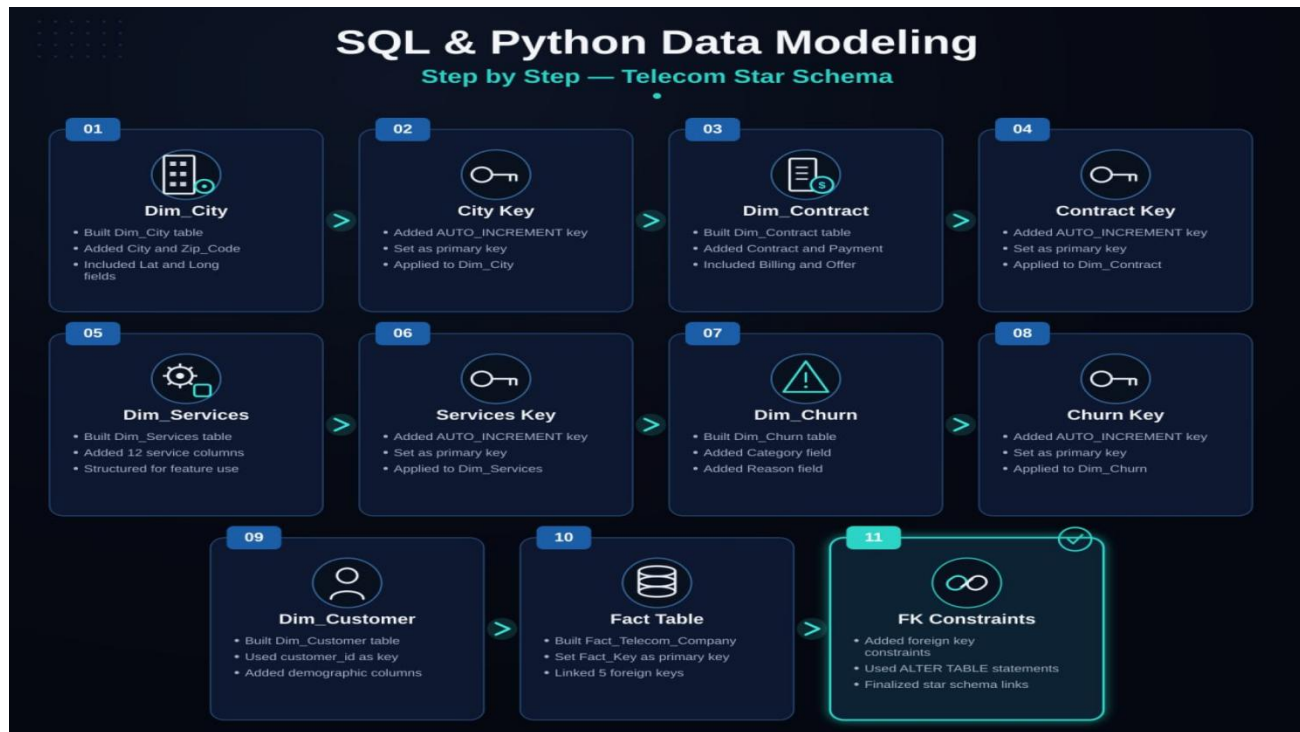
Fact_Telecom_Company: All foreign keys, measures (charges, revenue, tenure), and churn flag

Note on Data Modeling across tools:

In SQL (MySQL) and Python (SQLite3), the Star Schema was fully implemented with primary keys, foreign key constraints, and a 1:1 relationship between Dim_Customer and the Fact table enforced via a UNIQUE constraint on customer_id in the Fact table. The Fact table uses an independent surrogate primary key (Fact_Key / INTEGER PRIMARY KEY AUTOINCREMENT).

In Power BI and Tableau, the analysis was conducted directly on the Excel output files from the preprocessing phase, applying the same logical data model without recreating the full SQL schema.





2.5 Tools & Technologies



Microsoft Excel

A spreadsheet application used for data preprocessing via Power Query, building a Star Schema data model in Power Pivot, creating Pivot Tables, and developing interactive dashboards. Statistical analyses including T-Test, Chi-Square, ANOVA, and regression were performed using the Data Analysis ToolPak.

Power Query

Fact_Telecom_Customers - Power Query Editor

FileHomeTransformAdd ColumnView

Queries [7]

Churn

Dim_City

Dim_churn

Dim_Contract

Dim_Services

Fact_Telecom_Custo...

telecom_customer_ch...

Table.TransformColumnTypes(#Promoted Headers,{{"customer_id", type text}, {"City_Key", Int64.Type}, {"Contract_Key", Int64.Type},

	customer_id	City_Key	Contract_Key	Services_Key	churn_key	Gender	SeniorCitizen
1	0002-ORFBO	1	1	1	1	Female	Non-Senior
2	6689-VIRTK	9	1	105	1	Female	Senior
3	2673-CKQEU	148	7	105	1	Female	Senior
4	7590-VHVEG	2	2	2	1	Female	Non-Senior
5	0975-VOOVL	347	3	2	1	Female	Non-Senior
6	1078-TDCRN	347	8	2	1	Female	Senior
7	6549-YMFAW	29	1	23	1	Male	Senior
8	2682-KEVRP	31	1	23	1	Female	Senior
9	1237-WIYYZ	32	11	23	1	Female	Non-Senior
10	0030-FNXPP	47	3	23	1	Female	Non-Senior
11	0042-RLHYP	59	22	23	1	Female	Non-Senior
12	0048-PHNL	63	17	23	1	Female	Non-Senior
13	0040-HALCW	55	18	23	1	Male	Non-Senior
14	0354-VXWJC	64	18	23	1	Male	Non-Senior
15	7469-LKBCI	24	20	23	1	Male	Non-Senior
16	6563-VREX	38	25	23	1	Male	Non-Senior
17	1074-WVEVG	40	25	23	1	Female	Non-Senior
18	8191-XWSZG	34	26	23	1	Female	Non-Senior
19	1680-VDCVW	44	29	23	1	Male	Non-Senior
20	8665-UTDNZ	56	28	2	2	Male	Non-Senior
21	6317-YPKDH	333	36	2	2	Female	Non-Senior
22	1066-JKSGR	46	3	23	3	Male	Non-Senior
23	0031-PVLZI	49	30	23	4	Female	Non-Senior
24	0042-JVWQJ	57	33	23	1	Male	Non-Senior
25	1000-AJSLD	38	44	23	3	Male	Non-Senior
26	0052-YNYOT	67	7	23	1	Female	Non-Senior
27	7310-EGVHZ	68	38	23	1	Male	Non-Senior
28	0064-SLUDG	76	22	23	1	Female	Non-Senior
29	1029-OFFBN	80	69	23	3	Male	Non-Senior

Query Settings

PROPERTIES

Name

Fact_Telecom_Customers

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

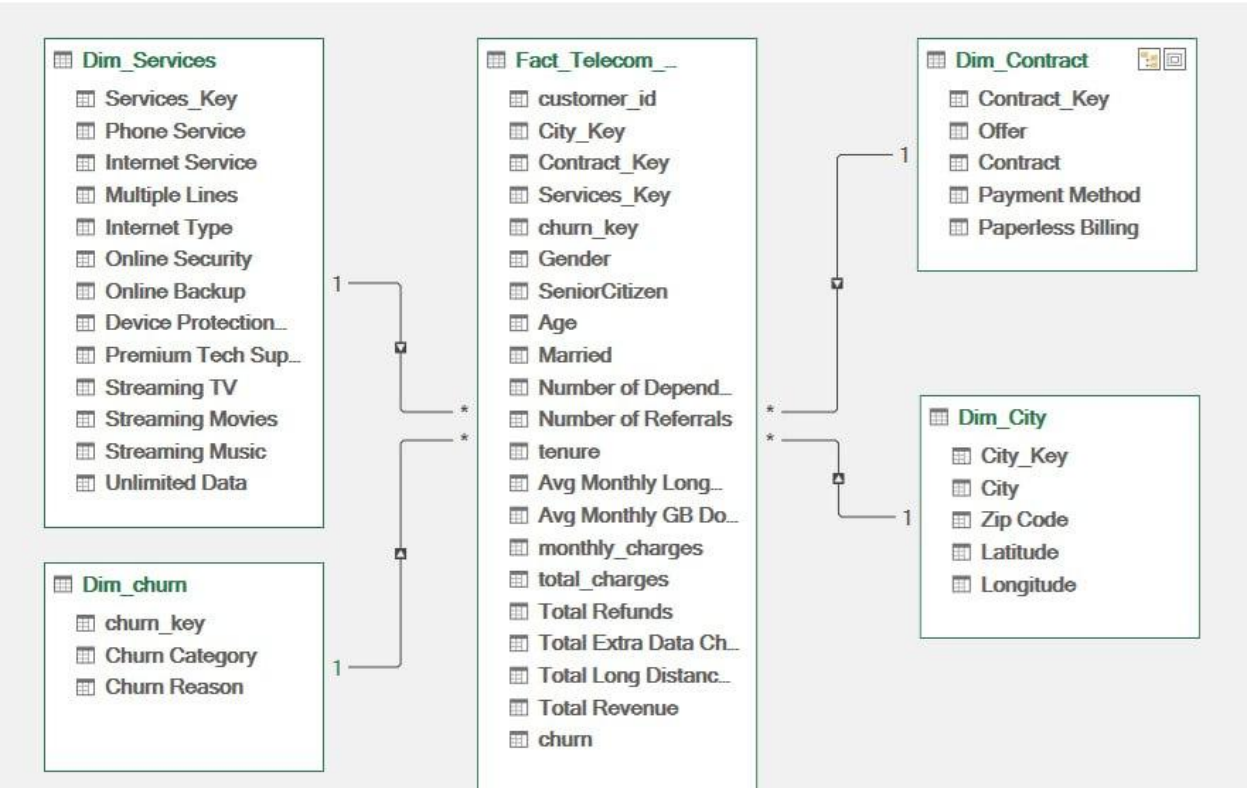
Changed Type

23 COLUMNS, 999+ ROWS

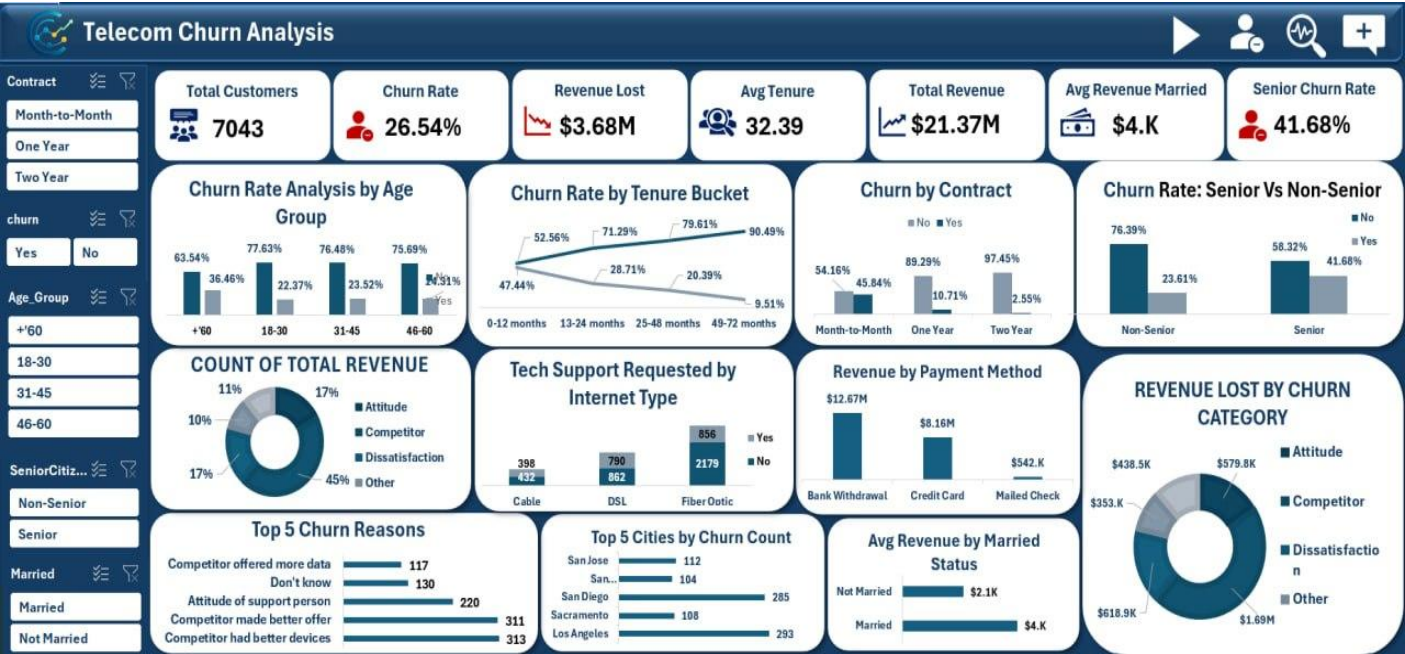
Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 00:00

Excel Schema



Excel Dashboard





MySQL (SQL)

A relational database management system used to implement the full data pipeline in SQL — including data cleaning, dataset merging, creation of Dimension and Fact tables, foreign key constraints, analytical queries, Views, and a Stored Procedure for churn risk scoring.

Clean Table Customer Churn

customer_id	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMov
7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No	No	No	No
5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No
3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No
7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes	Yes	No	No
9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No
9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes
1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	No
6713-QKQMC	Female	0	No	No	10	No	No phone service	DSL	Yes	No	No	No	No	No
7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes
6388-TABGU	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No

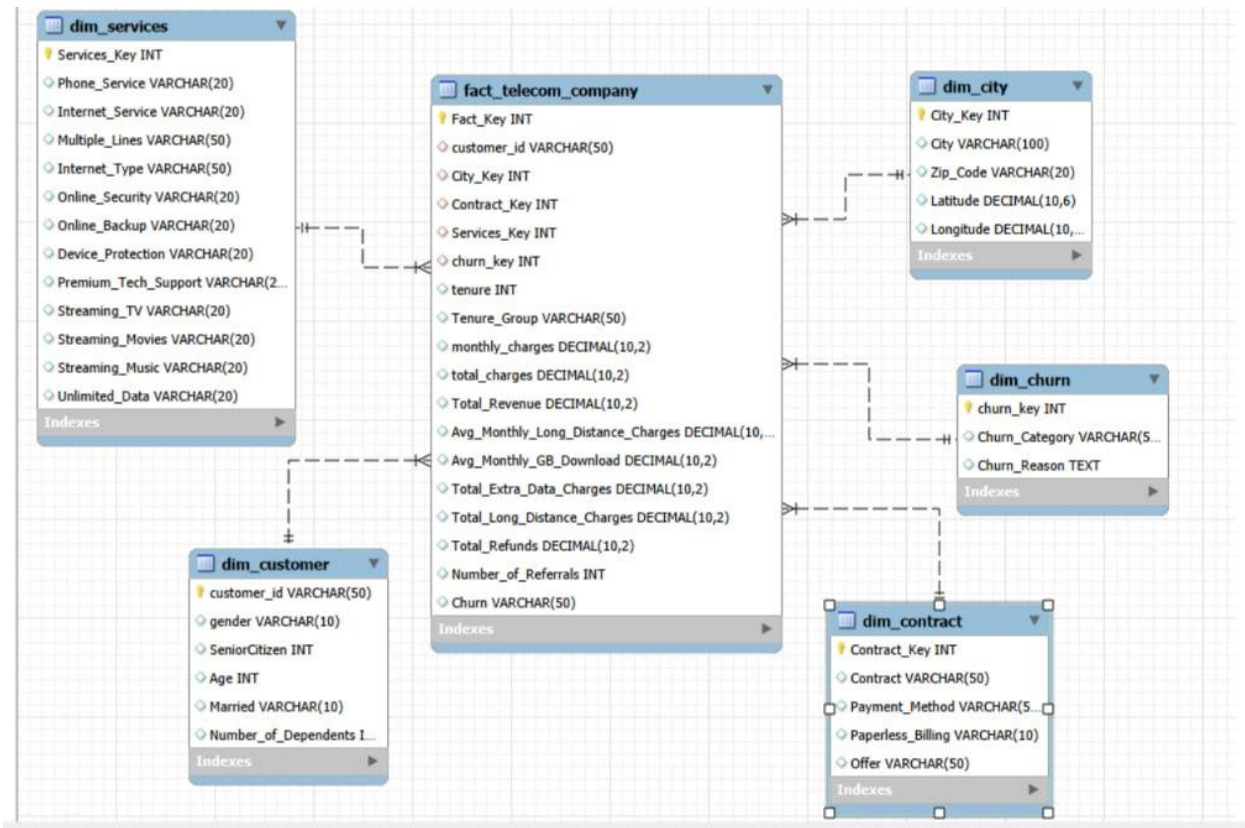
Clean Table Telecom Customer Churn

customer_id	gender	Age	Married	Number_of_Dependents	City	Zip_Code	Latitude	Longitude	Number_of_Referrals	tenure	Offer	Phone_Service	Avg_Monthly_Long_Distance_Ch
0002-ORFBO	Female	37	Yes	0	Frazier Park	93225	34.827662	-118.999073	2	9	None	Yes	42.39
0003-MKNFE	Male	46	No	0	Glendale	91206	34.162515	-118.203869	0	9	None	Yes	10.69
0004-TLHLJ	Male	50	No	0	Costa Mesa	92627	33.645672	-117.922613	0	4	Offer E	Yes	33.65
0011-IGKFF	Male	78	Yes	0	Martinez	94553	38.014457	-122.115432	1	13	Offer D	Yes	27.82
0013-EXCHZ	Female	75	Yes	0	Camarillo	93010	34.227846	-119.079903	3	3	None	Yes	7.38
0013-MHZWF	Female	23	No	3	Midpines	95345	37.581496	-119.972762	0	9	Offer E	Yes	16.77
0013-SMEOE	Female	67	Yes	0	Lompoc	93437	34.757477	-120.550507	1	71	Offer A	Yes	9.96
0014-BMAQU	Male	52	Yes	0	Napa	94558	38.489789	-122.270110	8	63	Offer B	Yes	12.96
0015-UOCOJ	Female	68	No	0	Simi Valley	93063	34.296813	-118.685703	0	7	Offer E	Yes	10.53
0016-QLJIS	Female	43	Yes	1	Sheridan	95681	38.984756	-121.345074	3	65	None	Yes	28.46

create Table Telecom Merged

Total_Extra_Data_Charges	Total_Long_Distance_Charges	Total_Refunds	Total_Revenue	churn	Churn_Category	Churn_Reason	SeniorCitizen	Tenure_Group
0.00	381.51	0.00	971.91	No	No churn	No churn	0	0-12 months
10.00	96.21	38.33	606.98	No	No churn	No churn	0	0-12 months
0.00	134.60	0.00	430.20	Yes	Competitor	Competitor had better devices	0	0-12 months
0.00	150.93	0.00	775.53	No	No churn	No churn	0	0-12 months
0.00	707.16	0.00	8495.86	No	No churn	No churn	1	49-72 months
20.00	816.48	0.00	6169.43	No	No churn	No churn	0	49-72 months
0.00	73.71	0.00	411.11	No	No churn	No churn	1	0-12 months
0.00	0.00	0.00	2440.80	No	No churn	No churn	0	49-72 months
0.00	1152.72	0.00	9562.32	No	No churn	No churn	0	49-72 months
0.00	162.00	0.00	7455.60	No	No churn	No churn	0	49-72 months

SQL Schema



Churn Rate: Senior vs Non-Senior

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	SeniorCitizen	Churn_Rate		
▶	0	23.61		
	1	41.68		

Tech Support Requested by Internet Type

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Internet_Type	Tech_Support_Yes	Tech_Support_No	
▶	Cable	398	432	
	DSL	790	862	
	Fiber Optic	856	2179	



Python

A general-purpose programming language used for data preprocessing with pandas, Star Schema implementation in SQLite3, exploratory data analysis, statistical testing, machine learning (Random Forest classifier with overfitting validation via 5-Fold Cross Validation), and building an interactive multi-page Streamlit dashboard.

Data Cleaning - Churn Table

Clean and standardize the churn dataset.

	customer_id	SeniorCitizen	tenure	gender	Partner	Dependents	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	Tech
0	7590-VHVEG	0	1	Female	Yes	No	No	No phone service	DSL	No	Yes	No	
1	5575-GNVDE	0	34	Male	No	No	Yes	No	DSL	Yes	No	Yes	
2	3668-QPYBK	0	2	Male	No	No	Yes	No	DSL	Yes	Yes	No	
3	7795-CFOCW	0	45	Male	No	No	No	No phone service	DSL	Yes	No	Yes	
4	9237-HQITU	0	2	Female	No	No	Yes	No	Fiber optic	No	No	No	

Data Cleaning - Telecom Customer Table

Clean customer, location, service, contract, payment, and churn columns.

	customer_id	gender	Age	Married	Number_of_Dependents	City	Zip_Code	Latitude	Longitude	Number_of_Referrals	tenure	Offer	Phone_Service	Avg_M
20	0023-UYUPN	Female	79	Yes	0	Ahwahnee	93601	37.38	-119.74	0	50	None	Yes	
21	0023-XUOPT	Female	26	Yes	0	Carnelian Bay	96140	39.23	-120.09	1	13	None	Yes	
22	0027-KWYKW	Female	30	Yes	2	Alpaugh	93201	35.87	-119.50	1	23	Offer D	Yes	
23	0030-FNXPP	Female	22	No	0	Keeler	93530	36.56	-117.96	0	3	None	Yes	
24	0031-PVLZI	Female	34	Yes	0	Korbel	95550	40.77	-123.80	0	4	Offer E	Yes	
25	0032-PGELS	Female	37	Yes	1	Palomar Mountain	92060	33.31	-116.82	1	1	None	No	
26	0036-IHMOT	Female	37	Yes	3	Parlier	93648	36.62	-119.52	10	55	None	Yes	

Merge Cleaned Data

Merge churn data with telecom customer data and prepare the final dataset.

	customer_id	gender	Age	Married	Number_of_Dependents	City	Zip_Code	Latitude	Longitude	Number_of_Referrals	tenure	Offer	Phone_Service	Avg_M
20	0023-UYUPN	Female	79	Married	0	Ahwahnee	93601	37.38	-119.74	0	50	None	Yes	
21	0023-XUOPT	Female	26	Married	0	Carnelian Bay	96140	39.23	-120.09	1	13	None	Yes	
22	0027-KWYKW	Female	30	Married	2	Alpaugh	93201	35.87	-119.50	1	23	Offer D	Yes	
23	0030-FNXPP	Female	22	Not Married	0	Keeler	93530	36.56	-117.96	0	3	None	Yes	
24	0031-PVLZI	Female	34	Married	0	Korbel	95550	40.77	-123.80	0	4	Offer E	Yes	
25	0032-PGELS	Female	37	Married	1	Palomar Mountain	92060	33.31	-116.82	1	1	None	No	

Exploratory Data Analysis

Charts and business insights from telecom churn data

	Tenure_Group	Churn_Rate	Retain_Rate
0	0-12 months	47.40	52.60
1	13-24 months	28.70	71.30
2	25-48 months	20.40	79.60
3	49-72 months	9.50	90.50



SQLite3 (Python Library)

A lightweight embedded database engine used within Python to implement the Star Schema as a persistent .db file, enabling SQL queries alongside pandas-based analysis in the same notebook environment.

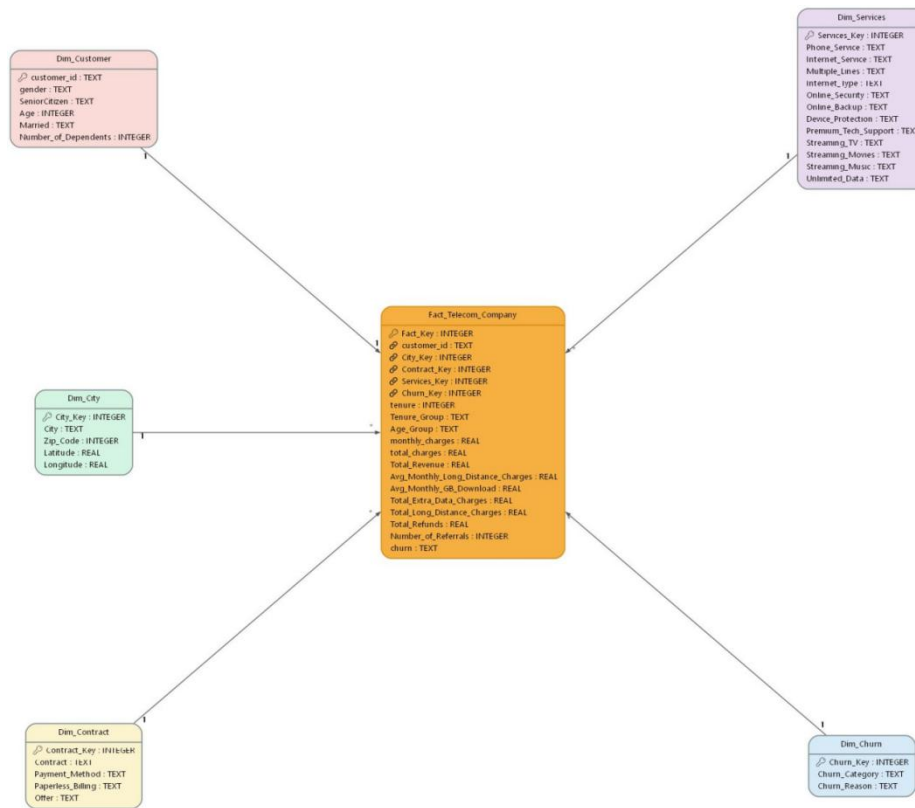
Dimension Tables

Create city, contract, services, churn, and customer dimension tables.

All Dimension Tables built:

```
Dim_City:      1,626 rows
Dim_Contract:  104 rows
Dim_Services:  966 rows
Dim_Churn:     21 rows
Dim_Customer:  7,043 rows
```

Python Schema (Graphviz)



Streamlit

An open-source Python framework used to build and deploy a multi-page interactive web application — "Telecom Churn Intelligence Platform" — featuring executive dashboards, customer segmentation views, ML prediction, churn risk scoring, and a What-If business strategy simulator.

Python Dashboard

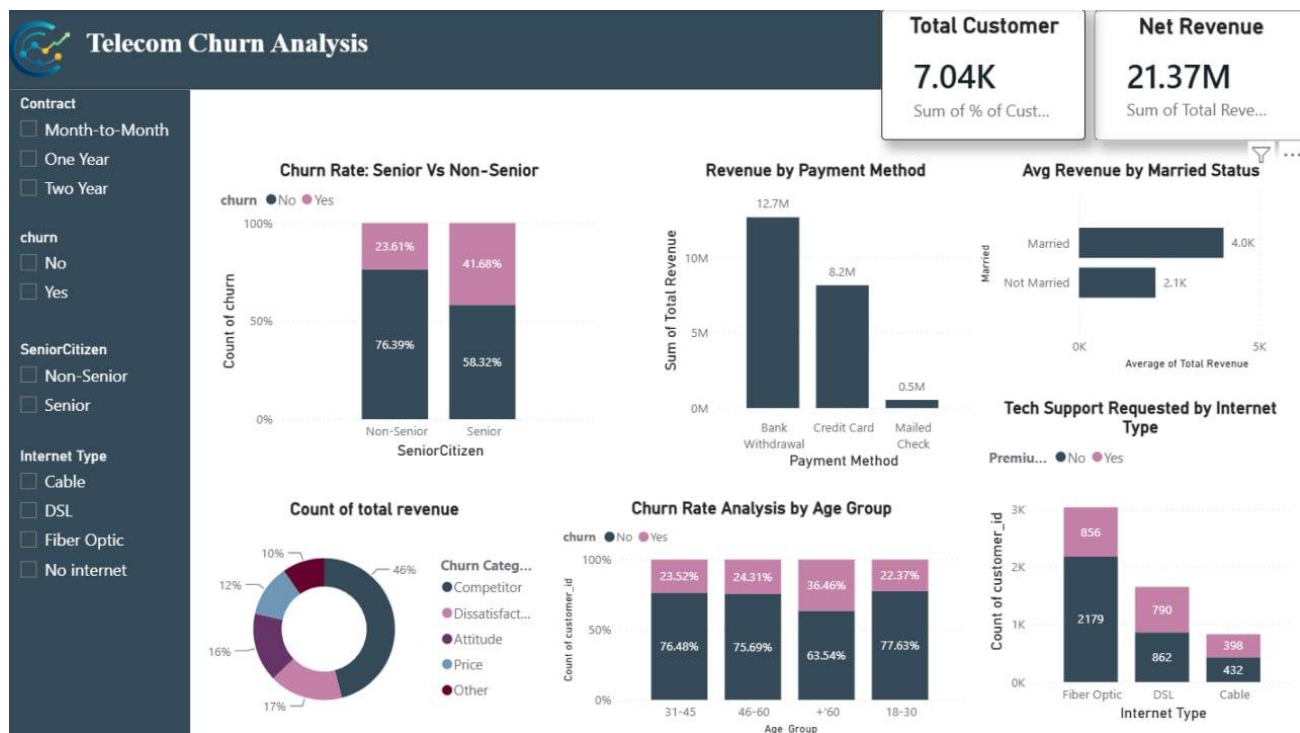




Power BI

A business intelligence platform used to build interactive dashboards with DAX measures, drill-through reports, and dynamic filtering. Power BI connected directly to the preprocessed Excel files and applied the same logical data model for analysis.

Power BI Dashboard



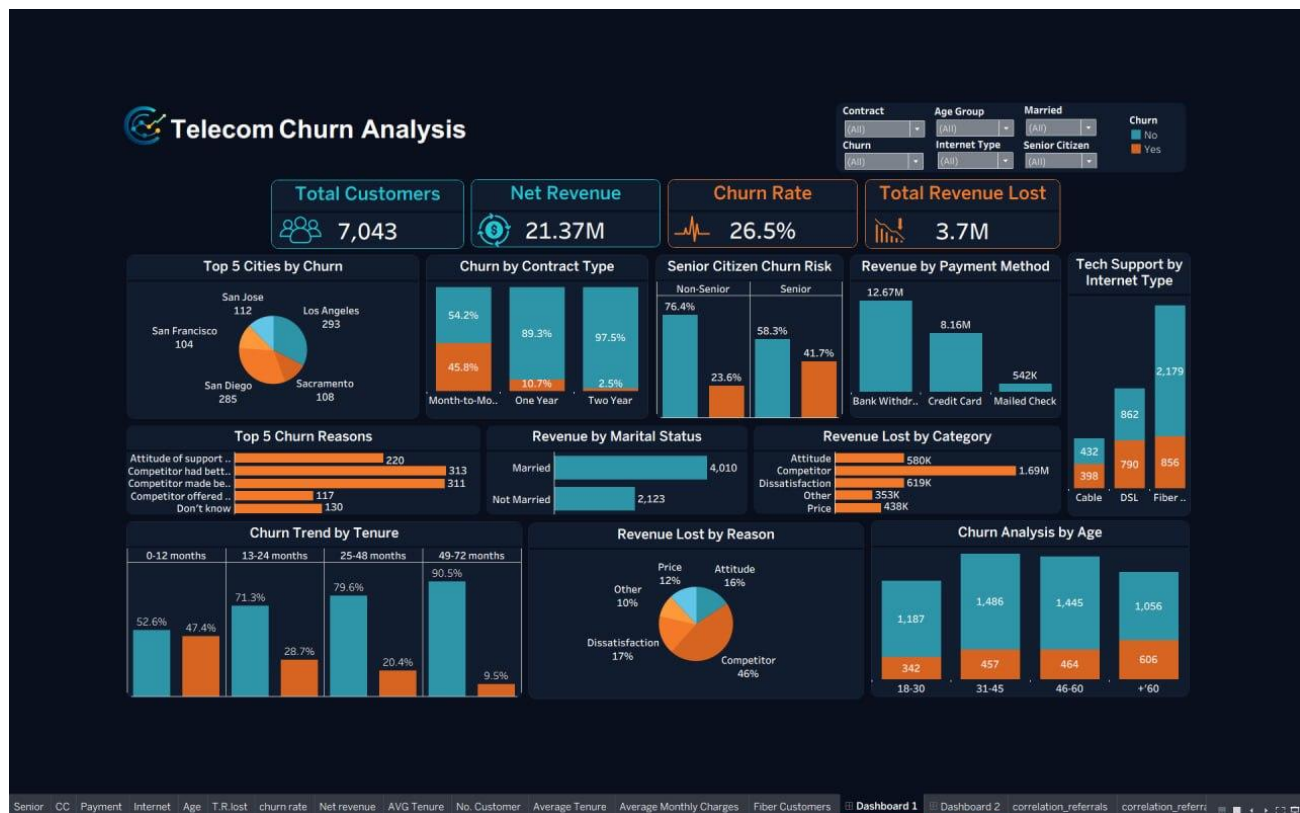




Tableau

A visual analytics platform used for geographic analysis, customer segmentation, and dashboard design. Tableau connected to the preprocessed Excel files and enabled visual exploration of churn patterns across cities, demographics, and service types.

Tableau Dashboard





Chapter 3

Insights & Recommendations

3.1 Key Insights

Insight 1 — The Loyalty Paradox

Churned customers paid an average of \$74 per month compared to \$61 for retained customers. This counter-intuitive finding reveals that the company is losing its premium-paying customers at a disproportionate rate — not its budget segment. Retention efforts must prioritize high-value customers, not just volume.

Insight 2 — The 12-Month Cliff

The churn rate in the first 12 months stands at 47.4%, dropping sharply to 28.7% in the second year and continuing to decline to 9.5% after four years. The first year of the customer relationship is by far the most critical — if a customer survives year one, the probability of long-term retention increases dramatically.

Insight 3 — The Security Effect

Customers without Online Security churn at 41.8% compared to only 14.6% for those with it — a nearly 3× difference. Online Security is not merely an add-on feature; it functions as a retention mechanism that increases service stickiness and reduces perceived switching ease.

Insight 4 — Senior Citizen Risk

Senior citizens (age 65+) churn at 41.7% compared to 23.6% for non-seniors. This demographic requires dedicated retention programs, as standard offers and digital-first communications may not adequately serve their needs.

Insight 5 — Competitor Intelligence Gap

45% of churned customers left for competitors — citing better devices (313 customers), better offers (311), better data allowances (117), and higher download speeds (100). No single competitor reason dominates, indicating that the company lacks structured intelligence on competitor positioning. An exit survey program is needed to close this gap.

Insight 6 — Contract Type as the Strongest Predictor

Month-to-Month contract customers churn at 45.8%, compared to 10.7% for One Year and just

2.5% for Two Year contracts. Contract type was also the most important feature in the Random Forest ML model. Long-term contract binding is the single most effective retention lever available.

3.2 Business Recommendations

Recommendation 1 — Launch a 6-Month Onboarding Program

Given that 47.4% of churn occurs in the first year, the company should implement a structured onboarding journey for all new customers. This includes welcome calls at day 7, service check-ins at month 3, and a loyalty offer at month 6. Reducing first-year churn by even 15% could prevent hundreds of cancellations monthly.

Recommendation 2 — Incentivize Contract Migration

Introduce targeted campaigns to migrate Month-to-Month customers to annual contracts, using loyalty discounts, service upgrades, or price-lock guarantees. Given the 43-point churn rate gap between contract types, this is the highest-ROI retention strategy available.

Recommendation 3 — Bundle Online Security by Default

Include Online Security in entry-level packages at no additional charge for the first six months. The service reduces churn by 3× and its cost is negligible compared to the revenue lost to a churned customer. This converts a retention tool into a default offering.

Recommendation 4 — Develop a Senior Citizen Retention Plan

Create a dedicated plan for senior customers featuring simplified service tiers, a dedicated support line, proactive outreach calls, and discounted pricing. With a 41.7% churn rate — nearly double the baseline — this segment represents a significant and addressable revenue risk.

Recommendation 5 — Implement Quarterly Competitive Pricing Reviews

Since 45% of churn is competitor-driven, the company should conduct formal competitive benchmarking every quarter, tracking pricing, device offers, and data allowances across major competitors. Findings should feed directly into product and pricing decisions.

Recommendation 6 — Deploy Structured Exit Surveys

Replace the current high-level churn reason categories with a structured exit survey capturing specific competitor offers, pricing comparisons, and service dissatisfaction details. This intelligence is essential for closing the competitor gap and enabling evidence-based pricing strategy.